
Location-based Services



Introduction

- ▶ Location-based services (LBS) are now popular and important: navigation, tourism, advertising,...
- ▶ To empower LBS, it is required to:
 - ▶ Display maps and information on it
 - ▶ Obtain knowledge about where the user is geographically located in real time

Maps

Maps Package (3rd Party)

- ▶ Installation

- ▶ `npm install react-native-maps`

- ▶ Attn: Web version is not available!

Showing a Map

```
import MapView, {PROVIDER_GOOGLE} from 'react-native-maps';

export default function App() {
  const initialRegion = {
    latitude: 21.006179,
    longitude: 105.8435599,
    latitudeDelta: 0.01,
    longitudeDelta: 0.01,
  };

  return <View style={...StyleSheet.absoluteFillObject}>
    <MapView style={...StyleSheet.absoluteFillObject}
      provider={PROVIDER_GOOGLE} // only for iOS
      initialRegion={initialRegion} />
  </View>
}
```

Capturing the Region Changes

▶ Events:

- ▶ `onRegionChange`: When the user is panning or zooming
- ▶ `onRegionChangeComplete`: When the user has finished panning or zooming

▶ Example:

```
const [region, setRegion]  
  useState(initialRegion);
```

```
<MapView ...  
  onRegionChange={r => setRegion(r)}  
  onRegionChangeComplete={r => setRegion(r)}  
>
```

Programmatically Moving to a Place

```
const mapRef = useRef(null);  
<MapView ref={mapRef} ...>  
  ...  
</MapView>  
  
<Button title="Go to Paris" onPress={() =>  
  mapRef.current.animateToRegion({  
    latitude: 48.856090,  
    longitude: 2.340084,  
    latitudeDelta: 0.2,  
    longitudeDelta: 0.2,  
  })  
} />
```

Showing Markers

```
import MapView, { Marker } from 'react-native-maps';

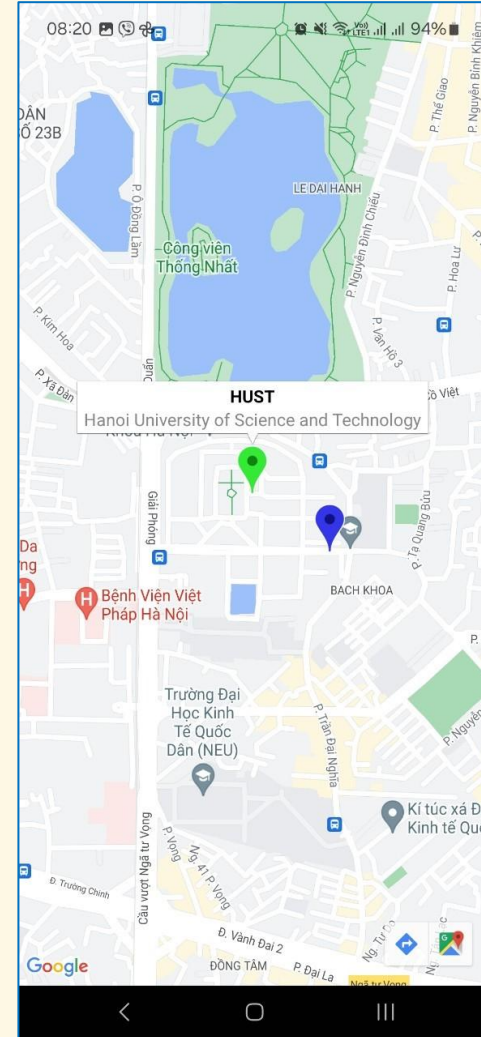
// ...

const hustPos = {latitude: 21.006179, longitude:
  105.8435599};

const gatePos = {latitude: 21.00501, longitude:
  105.8452214};

return <View style={styles.container}>
  <MapView style={styles.map}>
    <Marker coordinate={hustPos} pinColor="green"
      title="HUST"
      description="Hanoi University of Science and
      Technology" />
    <Marker coordinate={gatePos} pinColor="blue"
      title="HUST Gate" description="The gate of HUST" />
  </MapView>
</View>
```

- ▶ Attn: Make sure to put the <Marker> component as a child of <MapView>



Draggable Markers

▶ Add a state

```
▶ const [rv, setRV] = useState({  
    latitude: 21.006179,  
    longitude: 105.8435599  
});
```

▶ The marker

```
▶ <Marker draggable  
    onDrag={  
        e => setRV(e.nativeEvent.coordinate) }  
    onDragEnd={  
        e => setRV(e.nativeEvent.coordinate) }  
    coordinate={rv}  
    title="Rendez-vous" />
```

Overlays

- ▶ `<Circle>`, `<Polygon>`, and `<Polyline>` are available to pass in as children to the `<MapView>`

- ▶ **Example:**

- ▶

```
<Polygon tappable
  coordinates={hustBoundary}
  holes={holes}
  fillColor='#f007'
  strokeColor='#00f'
  strokeWidth={3}
  onPress={ (e) =>
    pressHandler(e.nativeEvent.coordinate) }
/>
```

Exercise

- ▶ Write an app that allows the user to add, remove, move markers.

Directions

Map Directions Package (3rd Party)

▶ Installation

▶ `npm install react-native-maps-directions`

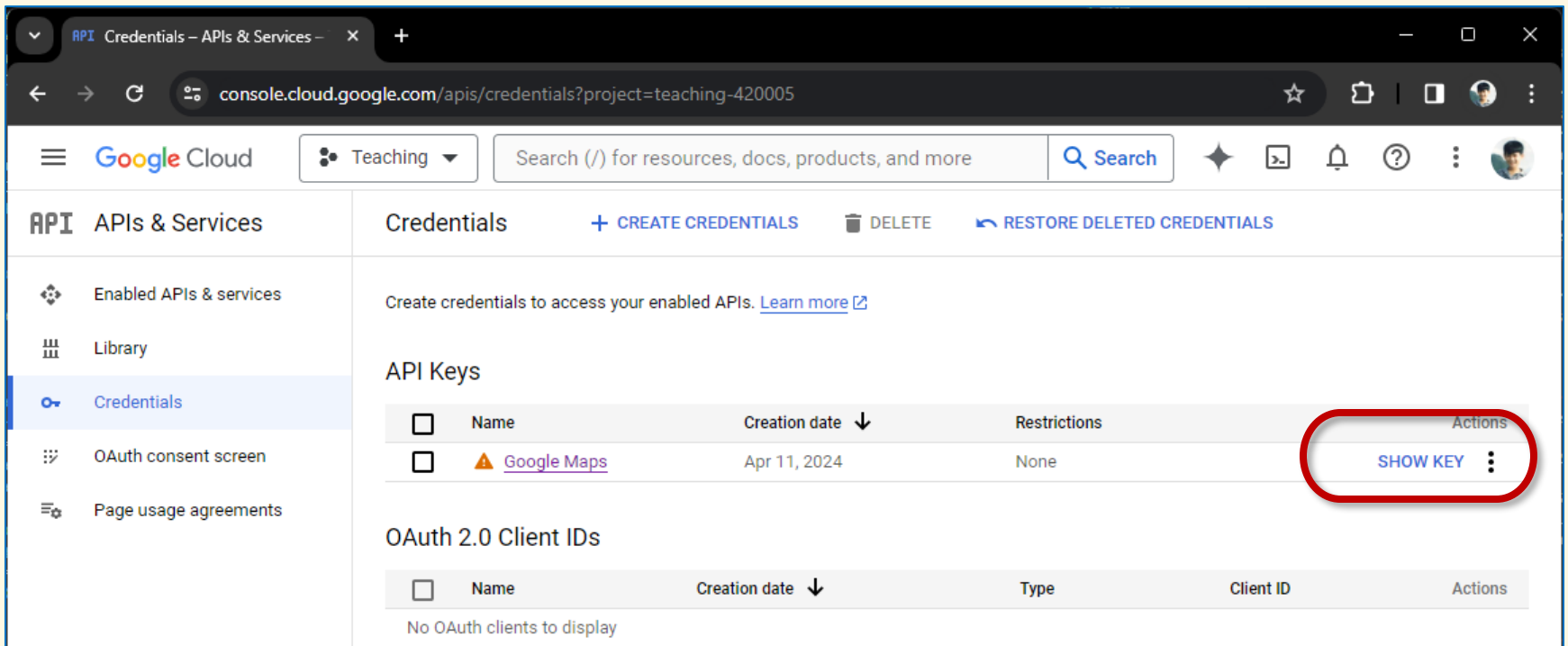
▶ Example:

▶ `const GOOGLE_MAPS_APIKEY = '...'; // see next slide`
`const origin = {`
 `latitude: 21.005089, longitude: 105.845635};`
`const destination = 'Hoan Kiem lake';`

▶ `<MapView ...>`
 `<MapViewDirections`
 `apiKey={GOOGLE_MAPS_APIKEY}`
 `origin={origin}`
 `destination={destination}`
 `mode='WALKING'`
 `strokeColor='#00fa'`
 `strokeWidth={5} />`
`</MapView>`

Creating a Free Google Cloud API Key

- ▶ Go to: <https://console.cloud.google.com/>
 - ▶ APIs and **Services** → **Credentials** → **Create Credentials** → **API key**
 - ▶ Press on **Show Key** to obtain the key
- ▶ Attn: Don't share the key to others!



The screenshot shows the Google Cloud Console interface. The left sidebar has a menu with 'APIs & Services' selected. The main content area is titled 'Credentials' and includes a '+ CREATE CREDENTIALS' button. Below this, the 'API Keys' section is displayed with a table:

☐	Name	Creation date ↓	Restrictions	Actions
☐	Google Maps	Apr 11, 2024	None	SHOW KEY ⋮

The 'SHOW KEY' button in the 'Actions' column for the 'Google Maps' entry is circled in red. Below the 'API Keys' section, the 'OAuth 2.0 Client IDs' section is visible, showing 'No OAuth clients to display'.

Exercise

- ▶ Write an app that allows the user to choose two places on a map, then show the route from the first to the second.

Geolocation

General

- ▶ React Native supports obtaining geolocation
 - ▶ On the web: Based on the web Geolocation API
 - ▶ On mobile devices: using a polyfill package
- ▶ Installation
 - ▶ `npm install expo-location`
 - ▶ For bare projects, on Android (AndroidManifest.xml file):
 - ▶

```
<uses-permission  
  android:name="android.permission.ACCESS_COARSE_LOCATION" />  
<uses-permission  
  android:name="android.permission.ACCESS_FINE_LOCATION" />
```
- ▶ On program startup
 - ▶

```
import * as Location from 'expo-location';  
Location.installWebGeolocationPolyfill();
```

Retrieving the User's Location (One Time)

```
const geolocationOptions = {
  enableHighAccuracy: true,
  timeout: 20000,
  maximumAge: 1000
}

const [loc, setLoc] = useState(null);

useEffect(() => {
  navigator.geolocation.getCurrentPosition(pos => {
    setLoc(pos.coords);
  },
  error => {
    // ...
  },
  geolocationOptions
), []);
```

Retrieving the User's Location Continuously

```
const geolocationOptions = {
  enableHighAccuracy: true,
  timeout: 20000,
  maximumAge: 1000,
  distanceFilter: 2
}

const [loc, setLoc] = useState(null);

useEffect(() => {
  navigator.geolocation.watchPosition(pos => {
    setLoc(pos.coords);
  },
  error => {
    // ...
  },
  geolocationOptions
), [])
```

Exercises

- ▶ Write an app that show the current user's location on a map.
- ▶ Write an app that show the user's traversed trajectory with a `<Polyline>`.